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B.Tech. DEGREE EXAMINATION, MAY 2023

Second Semester

21BTC101T - BIOCHEMISTRY

(For the candidates admitted from the academic year 2022-2023)

Note:

- (i) **Part - A** should be answered in OMR sheet within first 40 minutes and OMR sheet should be handed over to hall invigilator at the end of 40th minute.
- (ii) **Part - B** and **Part - C** should be answered in answer booklet.

Time: 3 Hours

Max. Marks: 75

PART – A (20 × 1 = 20Marks)

Answer ALL Questions

- | | Marks | BL | CO | PO |
|--|-------|----|----|-----|
| 1. A phosphodiester bond is present between
(A) Fatty acids in a diglyceride (B) Monosaccharides in carbohydrates
(C) Amino acids in proteins (D) Nucleotides in DNA | 1 | 2 | 1 | 1,4 |
| 2. Which of the following macromolecules have the general molecular formula (CH ₂ O) _n ?
(A) Proteins (B) Lipids
(C) Carbohydrates (D) Nucleic acids | 1 | 1 | 1 | 1,4 |
| 3. α-D-glucose and β-D-glucose are
(A) Epimers (B) Anomers
(C) Keto-ALDO pairs (D) Constitutional isomers | 1 | 1 | 1 | 1,4 |
| 4. The phospholipid that prevents the adherence of inner surface of lungs
(A) Phosphoinositides (B) Lecithin
(C) Cephalin (D) Plasmogens | 1 | 1 | 1 | 1,4 |
| 5. Which of the following has the highest standard reduction potential?
(A) NAD (B) FMN
(C) Oxygen (D) Pyruvate | 1 | 2 | 2 | 1,3 |
| 6. The following is the most preferred source of energy for the brain
(A) Glycogen (B) Glucose
(C) Fructose (D) Fatty acids | 1 | 2 | 2 | 1,3 |
| 7. Light independent reactions in photosynthesis takes place at
(A) Stromal matrix (B) Thylakoid lumen
(C) Photo system – I (D) Photo system – II | 1 | 1 | 2 | 1,3 |
| 8. The following uncouples electron transport in the mitochondrion from the production of chemical energy in the form of adenosine triphosphate (ATP)
(A) Thymine (B) Thermogenin
(C) Cytochromes (D) Heme | 1 | 1 | 2 | 1,3 |

9. Which of the following are the major sites for glycogen storage
 (A) Muscle and brain (B) Muscle and liver
 (C) Liver and kidney (D) Adipose tissue
10. Which of the following is the precursor of glycogen?
 (A) UDP glucose (B) Malate
 (C) Glycerol 3-phosphate (D) Leucine and lysine
11. Which of the following hormones maintain blood glucose level by activation of gluconeogenesis
 (A) Epinephrine (B) Insulin
 (C) Glucagon (D) Nor-epinephrine
12. The following reactions generate acetyl CoA from pyruvate
 (A) Oxidative decarboxylation (B) Reduction
 (C) Dehydration (D) Dephosphorylation
13. One of the following is NOT an amino acid
 (A) Alanine (B) Proline
 (C) Choline (D) Glycine
14. The amino acids are said to be ketogenic when the carbon skeleton is finally degraded to
 (A) Fumarate (B) Acetyl CoA
 (C) Pyruvate (D) Succinyl CoA
15. The following amino acid do not participate in transamination
 (A) Glutamate (B) Tryptophan
 (C) Alanine (D) Lysine
16. The common compound shared by both TCA cycle and urea cycle is
 (A) α -keto glutarate (B) Fumarate
 (C) Oxalo acetate (D) Succinyl CoA
17. The following is the source of all carbon atoms in cholesterol
 (A) Acetyl CoA (B) Propionyl CoA
 (C) Malonyl CoA (D) Acetoacetyl CoA
18. The parent nucleotide synthesized in the biosynthesis of purines is
 (A) Orotidine monophosphate (B) Adenyl succinate
 (C) Xanthosine monophosphate (D) Inosine monophosphate
19. The amino acids required for the synthesis of pyrimidines are
 (A) Aspartate and glutamine (B) Alanine and cysteine
 (C) Asparagine and leucine (D) Methionine and tyrosine
20. Ketone bodies are formed from
 (A) Acetyl CoA (B) Acetone
 (C) Acetoacetate (D) Propionyl CoA

PART – B (5 × 8 = 40 Marks)Answer **ALL** Questions

Marks BL CO PO

21. a. What are enzymes? Draw the activation energy diagram with and without the enzyme and describe the major classes of enzymes. 8 2,3 1 1,4

(OR)

- b. What is pH? Define Bronsted-Lowry acid and base and derive the Henderson-Hasselbalch equation. 8 2,3 1 1,4
22. a. Give an account on photorespiration and explain how C4 plants avoid photorespiration. 8 2,3 2 1,3

(OR)

- b. What is shuttle pathway? Explain one of the shuttle pathways in detail. 8 2,3 2 1,3
23. a. What is diabetes mellitus? Describe briefly about the blood glucose level regulation by insulin. 8 2 3 1,4,5

(OR)

- b. What is glycogenolysis? Which organs carryout this process? Briefly describe about glycogenolysis. 8 2,3 3 1,4,5
24. a. Write the salient features of transamination of amino acids and describe the mechanism of transamination in detail. 8 2 4 1,2,4

(OR)

- b. Mention any four amino acid metabolic disorders and describe about phenylketonuria in detail. 8 2 4 1,2,4
25. a. Discuss the steps involved in the β -oxidation of fatty acid. 8 2 5 1,2

(OR)

- b. Explain the process by which pyrimidines are synthesized. 8 2 5 1,2

PART – C (1 × 15 = 15 Marks)Answer **ANY ONE** Questions

Marks BL CO P

26. Explain the electron transport chain with the help of a diagram and discuss the synthesis of ATP by oxidative phosphorylation with the help of chemiosmotic theory / hypothesis proposed by Peter Mitchell. 15 4 2
27. Ammonia is a toxic product of nitrogen (amino acid) metabolism and it must be removed from the body. Explain the mechanism of removal of ammonia from body. 15 4 4